BMC Public Health

SYSTEMATIC REVIEW Open Access

Leveling up immunization: a systematic review of gamification interventions to improve Human Papillomavirus (HPV) vaccine uptake

Shafaq Taseen[1,3*], Ramsha Ayub[1], Gul E Lala Noman[2] and Mohammad Tahir Yousafzai[1,4]

Abstract

Background HPV vaccine is effective in preventing cervical cancer in adolescents and young adults. Despite its proven effectiveness, vaccination rates remain suboptimal due to several barriers, such as insufficient knowledge and vaccine hesitancy. Gamification has emerged as an effective strategy for getting people involved and altering their health-related behaviors. To date, there are no literature-based systematic reviews on the effectiveness of gamification in leveling up HPV immunization and knowledge among adolescents or young adults. This systematic review seeks to evaluate the impact of gamification interventions on HPV vaccine knowledge, intentions, and attitudes in adolescents and young adults.

Method This systematic review was conducted using PRISMA guidelines and PI(E)COS model. The study protocol was registered in PROSPERO: CRD42024557308. Peer-reviewed scientific articles regarding gamification interventions in the context of promoting HPV were included. The database search encompassed PubMed, Scopus, Cochrane Central, PsycINFO, as well as manual searches from Google Scholar.

Results This systematic review identified 216 records, of which nine studies met the full inclusion criteria. These studies focused on adolescents and young adults between the ages of 11 and 26 across different geographical locations. The reviewed studies indicate that gamification interventions, particularly those delivered through digital platforms, can effectively enhance knowledge, positively influence attitudes, and improve intentions to receive the HPV vaccine among adolescents. Customization features and interactive elements in the gamified intervention played a crucial role in engaging participants and driving vaccine-related behaviors. However, the findings were not universally consistent, with some studies reporting no significant impact.

Conclusion The evidence from this review supports the potential of gamification interventions to positively influence adolescent health, particularly in the context of HPV vaccination. Nonetheless, the review is limited by the small number of quantitative studies. Consequently, larger-scale studies are necessary to validate the effectiveness of these interventions robustly.

*Correspondence:
Shafaq Taseen
s.angilahasan@gmail.com

Full list of author information is available at the end of the article

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Prospero registration CRD42024557308.

Keywords Gamification, HPV vaccination, Human papillomavirus, Vaccine uptake, Adolescents, Vaccine behaviour, Serious games, Game-Based learning

Background

Human papillomavirus (HPV) stands as one of the most prevalent sexually transmitted infections (STIs) affecting adolescents and young adults, as reported by the World Health Organization [1]. This makes HPV a pervasive public health challenge with a staggering emergence of approximately 14 million new cases every year, a global burden of HPV, as per the recent 2023 reports of *Global HPV Statistics reports* [2] is presented in Fig. 1. An estimated 80 million individuals currently harbor HPV infections within the United States [3]. Being a DNA tumor virus, HPV spreads easily through both cutaneous and mucous surfaces due to its ability to induce changes in epithelial proliferation [4]. HPV-associated cancers pose a significant health human health either for both males and females nowadays [5].

Multiple studies have supported the effectiveness of HPV vaccination as a prevention strategy. The HPV 9 quadrivalent vaccine Gardasil and a bivalent vaccine Cervarix have shown a remarkable efficacy, preventing approximately 90% of cervical and anal cancers, around 70% of oropharyngeal cancers, approximately 75% vaginal cancers, about 69% vulvar cancers, about 63% penile cancers, and preventing (close to 90%) genital warts [5, 6]. Cohort studies have observed a steep decline in the incidence of anogenital warts, with reductions of up to 45% among vaccinated populations [7]. Despite the demonstrated effectiveness of HPV vaccination, the vaccinated population is still not sufficient in numbers [8]. The unvaccinated population is still at high risk for HPV-related diseases. The Centers for Disease Control and Prevention (CDC), American Cancer Society, and American Academy of Pediatrics recommend initiation of the HPV vaccination series by the age of 13 to prevent HPV-related cancers and related morbidity in both males and females [9]. Yet, vaccination against HPV is significantly lagging behind other adolescent vaccines, such as Tdap and meningococcal vaccines. Most states have less than 50% of their population up-to-date with the HPV vaccine, which is far from the recommended 80% goal for Healthy People 2030 [10, 11]. Barriers to HPV vaccination exist at multiple levels, including among providers, parents, and interpersonal levels. Additional barriers include infrequent patient reminders about vaccination schedules, low clinician awareness of eligible youth for vaccination (i.e., missed opportunities), lack of parental knowledge on dosing and schedule, and insufficient patient-provider communication on vaccine hesitancy. As illustrated in Fig. 1, HPV incidence and mortality are disproportionately higher in low- and middle-income countries, where vaccination coverage remains suboptimal. These disparities underscore the need for innovative, accessible, and engaging interventions to increase vaccine uptake.

Gamification, an innovative approach that incorporates game design elements into non-game contexts, offers a promising strategy to engage individuals and modify behaviors [12]. There are many aspects of this multidimensional concept, such as fun interfaces, instant feedback, reward systems, challenges, and games [12, 13] playing in teams [14], avatars or quizzes [15]. A wide range of research has shown that gamification is effective in promoting positive health-related behaviors [12–15]. Gamification has the potential to enhance key predictors of vaccine uptake, including knowledge, attitudes, beliefs, behaviors, and vaccination intention [16]. These factors directly address well-documented barriers to vaccination, such as misinformation, low risk perception, vaccine hesitancy, and lack of motivation to complete vaccination schedules, which positively influence behavior change and improve vaccination-related knowledge as per the findings of Montagni et al. [17]. Despite the large number of primary studies, to date, there has been no systematic review on gamification interventions for HPV vaccination uptake. This gap in the literature indicates a need for an overarching synthesis of existing research.

While the benefits of gamification have been demonstrated across health domains, evidence specific to HPV vaccination remains fragmented and reported in diverse formats. Most existing studies present gamification as a bundled intervention without disentangling which components (e.g., feedback, avatars, leaderboards, rewards) drive outcomes. Clarifying these mechanisms is essential for practice, as public health practitioners and developers need evidence on which elements to prioritize when designing scalable and culturally appropriate interventions. A systematic review is therefore warranted, as it not only synthesizes heterogeneous evidence across settings but also provides a transparent framework for evaluating the contribution of individual components. This approach addresses a critical gap in the literature by moving beyond whether gamification works to understanding how and why specific elements influence HPV vaccine-related knowledge, attitudes, and uptake.

Study question

Do gamification interventions lead to increased HPV vaccine-related knowledge, positive changes in attitudes or

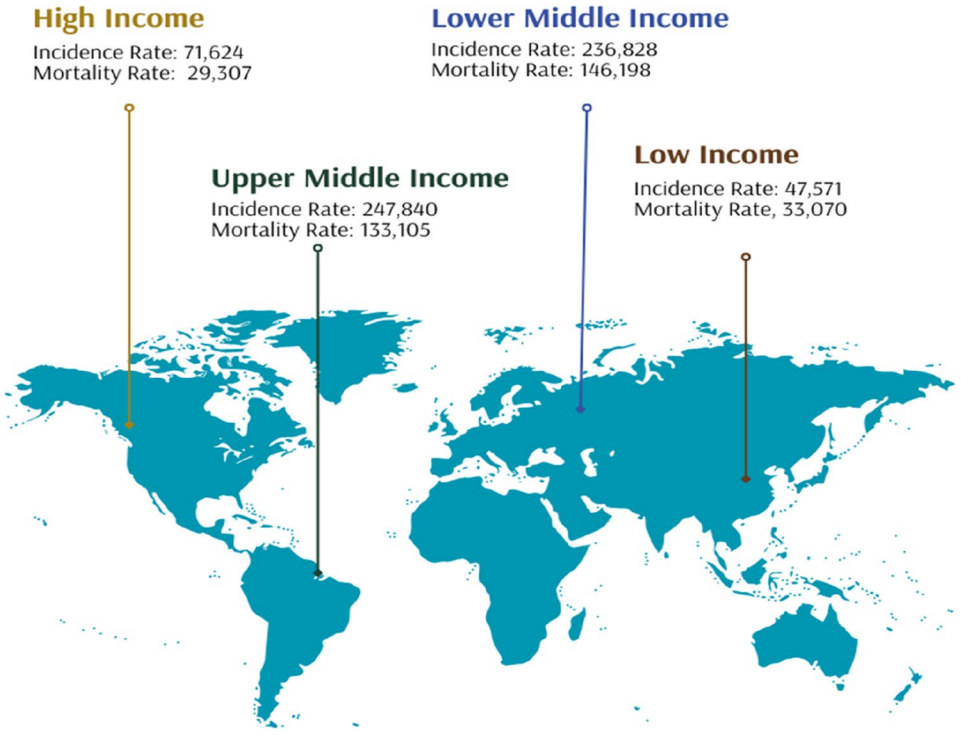


Fig. 1 Global burden of HPV-related cancer (2023 HPV Information Centre report)

behaviors related to HPV vaccination, and enhance HPV vaccine uptake rates or uptake intention among adolescents and young adults eligible for HPV vaccination?

1. Do gamification interventions improve HPV vaccine-related knowledge among adolescents and young adults?
2. Do gamification interventions promote positive changes in attitudes and behaviors toward HPV vaccination among adolescents and young adults?
3. Do gamification interventions enhance HPV vaccine uptake or intention to vaccinate among adolescents and young adults?
4. Do specific gamification components and mechanics influence HPV vaccine-related knowledge, attitudes, behaviors, and vaccination intentions among adolescents and young adults?

Study aims and objectives

The objectives of this study are.

a) To investigate the changes in HPV vaccine-related knowledge after gamification intervention among adolescents and young adults.
b) To investigate the changes in attitudes and behaviours related to HPV vaccination after gamification intervention among adolescents and young adults.
c) To evaluate the changes in HPV vaccine uptake rates or uptake intention after gamification intervention among adolescents and young adults.
d) To examine the role of specific gamification components and mechanics in enhancing HPV vaccine knowledge, vaccination-related behavior, attitudes, and vaccination intention among adolescents and young adults.

Methods

Systematic review protocol

To achieve an efficient and standardized plan for this systematic review, it was conducted according to the Population, Intervention/Exposure, Comparison, Outcome, and Setting (PI(E)COS) model and the PRISMA-P 2020 checklist [18]. The current systematic review protocol proposal was registered at the International Prospective Register of Systematic Reviews (PROSPERO) *CRD42024557308* to enhance transparency and inter-study reporting.

PI(E)COS elements

Population: Adolescent and young adults aged between 11 and 26 years, aligning the age range with WHO recommendations (targeting adolescents aged 9–14 and catch-up in young adults).

Intervention/Exposure: Gamification interventions.

Comparison: None.

Outcome: HPV vaccine uptake rates or uptake intention, Knowledge related to HPV vaccination and Attitudes and behaviors related to HPV vaccination.

Setting: Global Healthcare settings and community-based programs (both clinical e.g., hospitals, clinics, vaccination centers and non-clinical settings e.g., schools, universities, community outreach programs without any geographical restrictions).

Eligibility criteria

Studies were eligible for inclusion if they reported measurable outcome data related to HPV vaccination uptake, intentions, knowledge, or attitudes/behaviors following a gamification intervention. We considered all relevant study designs, including randomized controlled trials (RCTs), quasi-experimental studies, pre–post intervention studies, cross-sectional studies, and qualitative designs that evaluated gamification interventions targeting HPV vaccine knowledge, attitudes/behaviors, or vaccination intentions. Only studies published in peer-reviewed journals between 2000 and 2024, with full text available in English, were included in this systematic review.

Exclusion criteria

Studies not focusing on gamification interventions or with insufficient data or those not meeting the specified eligibility criteria were excluded. Review articles, LTE/editorials, and publications with incomplete data and participants outside the age range of adolescents and young adults eligible for HPV vaccination were also excluded.

Identification of relevant studies

A comprehensive systematic electronic search was conducted for studies published between January 2000 and September 2024. This search was performed independently in key proceedings-level databases, including PubMed, SCOPUS, Cochrane Central Register of Controlled Trials, and PsycINFO, using relevant keywords and Medical Subject Headings (MeSH) terms. Additionally, to enhance the search, a manual search of Google Scholar was undertaken to identify additional peer-reviewed studies. This extensive search strategy ensured that our review incorporated a diverse range of potentially eligible studies.

Search strategy

A systematic search strategy was employed to identify relevant articles, utilizing Boolean operators "OR" and "AND" to capture synonyms and combine concepts of Gamification interventions and HPV vaccine uptake, adherence, respectively. MeSH subheadings incorporated where applicable to enhance search precision, to account for variations in word forms and spellings, truncation and wildcards was employed such as ("gamification" OR "game design" OR "game-based" OR "interactive") AND ("HPV vaccine" OR "human papillomavirus vaccination" OR "cervical cancer prevention") OR HPV AND ("serious game*" OR "video game*" OR "therapeutic game*" OR "online game*" OR "game*" OR "game app*" OR "mobile game*" OR "digital game*" OR "gamif*"). The search strategy was tailored to each database, incorporating terms related to the associations defined by PICOS and modified search terms combined with MeSH terms. Details of the search strategy being used in each database are present in Supplementary File 1.

Study selection

Records were initially screened by title and abstract to remove those not relevant to the study and multiple reports from the same study. Full-text articles were then assessed for inclusion based on the eligibility criteria. Full text of potentially eligible records retrieved and reviewed. Both title/abstract screening and full-text review were conducted independently by two reviewers. Any disagreements were first discussed, and if consensus could not be reached, they were resolved by a third reviewer. Data extraction was also done independently by two reviewers using a standardized data extraction form Table 1. Extracted data include: study characteristics (e.g., author, year, study design), geographical data, participant age, and gender. Detailed coding of the intervention components for each study is provided in Supplementary File 2.

Data synthesis

A narrative synthesis of findings was done to summarize the effect of gamification interventions on HPV vaccine uptake rates, adherence to vaccination schedules, knowledge, attitudes, and behaviors towards vaccination. A narrative synthesis was conducted due to the high heterogeneity across the included studies in terms of design (qualitative and quantitative), intervention type, outcome measures, and reporting formats, which made a meta-analysis inappropriate. To ensure methodological rigor, risk of bias was assessed across five domains using the Cochrane RoB-2 tool for randomized trials and ROBINS-I for non-randomized/qualitative studies. The review followed the PRISMA 2020 checklist provided in Supplementary File 3 to ensure transparent reporting.

Results

Identification of studies

The flow diagram presents the selection process for studies included in this systematic review, shown in Fig. 2.

A total of 216 records were identified from various databases, including PubMed ($n = 95$), Scopus ($n = 61$),

Table 1 Study characteristics of included studies on HPV and gamifications

Authors	Geographical Context	Study design	Target population	Participants group	Sample Size	Age of Participants	Sex of Participants
(Cates, Crandell, et al., 2018)[19]	North Carolina	Non-Randomized Controlled Trial	Preteens in North Carolina	3 groups Focus Group 1: 7 participants Focus Group 2: 2 participants Focus Group 3: 7 participants	16	11–12 years	Male and Female
Cates et al., 2018 [24].	North Carolina	Qualitative- FGD	Preteens in North Carolina	2 groups Intervention arm: 19,398 Comparison group: 127,896	147,294	11–12 years	Male and Female
Cates et al., 2020 [23]	USA	Randomized Controlled Trial (RCT)	Participants who visited clinical sites	2 groups Intervention arm: 28 preteens Comparison arm: 27 preteens	47	11–13 years	Male and Female
Darville et al., 2018 [25]	USA	Randomized Controlled Trial (RCT)	University students	4 groups 4 groups: 108 participants	108	18–26 years	Male
Epstein et al., 2021[22]	Australia	Randomized Controlled Trial (RCT)	31 schools	19 schools Experimental arm: 7 schools) Control arm: 12 schools	3087	11–14 years	Male and Female
Occa et al., 2022 [26],	Italy	Mixed-methods FGD and experiment.	Middle school children	Watched animated game: 20 participants Played game: 15 participants	35	11–12 years	Male and Female
Ruiz-Lopez et al., 2019 [27]	Norway	Qualitative- FGD	High school students	2 groups Focus Group 1: 4 participants Focus Group 2: 22 participants	26	16–18 years	Males and Females
Hasni et al., 2023 [28]	Malaysia	Mixed-methods FGD and cross-sectional	University students	No groups	9	8–17 years	Male and Female
Chau et al., 2023 [21]	Hong Kong	Qualitative-Interview	Secondary School students of grades 8–11	8 Dyads 8 mother-daughter dyads	26	14–17 years	Females

Cochrane Library ($n=10$), PsycINFO ($n=15$), and manually from Google Scholar ($n=35$). After removing duplicates ($n=55$) and records not in English ($n=5$), 141 records remained for screening. 141 Records were screened based on their abstract and title. After this screening, 118 reports were sought for retrieval.

A further exclusion was made based on specific criteria: 15 review articles were excluded. 4 conference papers, 3 books, and 1 LTE/Case report were also excluded. Out of the 118 reports retrieved, 26 were assessed for eligibility. Further exclusions were made: 12 reports were excluded because they lacked the required data, 2 reports were excluded for not detailing the game interventions, and 2 reports were excluded as they were incomplete randomized controlled trials (RCTs). 1 report was excluded due to a lack of full access. Finally, 9 studies were included in the systematic review after fulfilling all the eligibility criteria.

Overall, most studies showed some concerns, with high risk in non-randomized designs [19–21] while only [22, 23] demonstrated lower risk, highlighting the need for more rigorous trials in this area. A traffic-light plot was generated in Python to visually summarize the overall judgments for each included study is provided in Supplementary File 1.

Study characteristics

Four studies were conducted in the USA [19, 23–25]. Single study from Australia [22]. Two studies took place in Europe, specifically in Italy [26] and Norway [27]. Two studies were conducted in Asian regions, one in Malaysia [28] and one in Hong Kong [21].

Four studies evaluated interventions, with three employing a randomized controlled trial (RCT) design [19, 22, 23, 25] and one utilizing a non-randomized controlled trial design [19], highlighting a focus on evaluating intervention effectiveness in controlled environments. Three studies employed qualitative methods [21, 24, 27], with one relying on interviews and the others on focus group discussions (FGDs). Two studies adopted mixed-methods approaches, primarily involving FGDs, presented in Table 1. Notably, the studies [22, 24] featured large sample sizes of 147,294 and 3,087 participants, respectively, whereas the smallest sample size observed in [19, 23–25], with 16 and 9 participants, focused on a qualitative approach. Most studies focused on preteens and adolescents aged 11 to 14 years [27] and [21] included participants aged 14 to 18 years [25] and [28] involved university students, with participants ranging from 18 to 26 years old.

Seven out of nine studies included both male and female participants, ensuring balanced gender representation. While Study [21] specifically targeted female

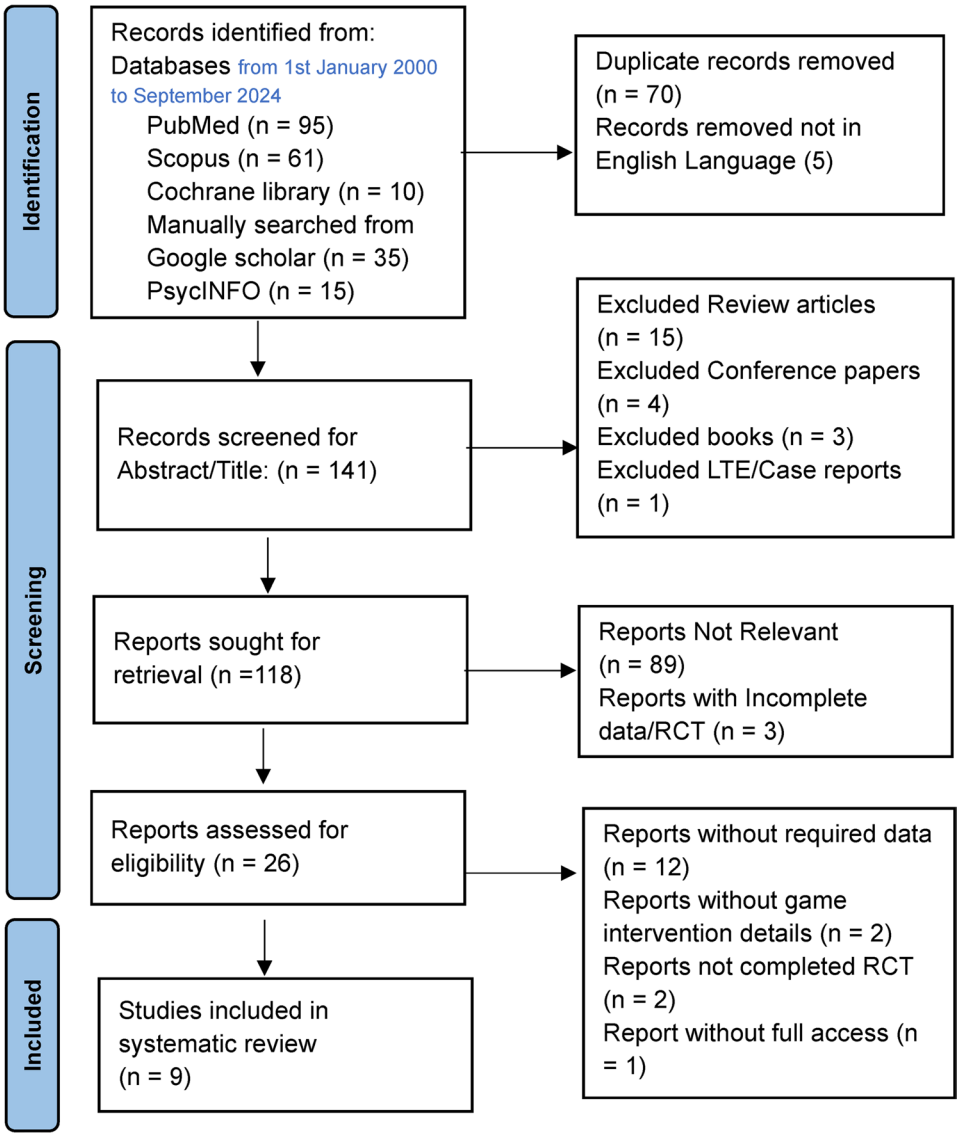


Fig. 2 PRISMA Flowchart diagram

participants, particularly mother-daughter dyads, and [25]focused on male HPV vaccination.

Type of gamification intervention and platform used

All studies employed a digital platform except Epstein [22], who employed a physical educational card game called "*Vaxcards*". Some studies used a web-based video game for a digital gamification intervention [19, 23–25]. Some implemented web-based puzzle games [25, 28]., while [27] Utilized a mobile app game [21]. Incorporated a digital game "*Kahoot*" and included animated videos. Seven of these studies focused on serious games, except [26], which combined a multimedia quiz with animated videos in a digital format.

Employed game mechanics

Game mechanics varied across studies. Some studies [25] featured customizable avatars to navigate HPV-related scenarios, and used puzzle-based levels with a leaderboard system [19, 23–25, 27]. Employed a web-based video game, '*Land of Secret Gardens*', that has a garden metaphor to promote HPV vaccination [26], games '*Salute e HPV* (Health and HPV) was designed using a human-centered approach to address the needs of Italian middle school children. It is a web-based quiz that can be played individually at home or in groups at school. In this study, children played in pairs due to limited computer availability while [28] combined obstacle navigation and virus shooting with HPV-related questions to advance game levels [22]. Used collectible card game '*Vaxcards*'

with characters based on diseases that children are vaccinated against.

Role of gamification intervention

Three studies [19, 23, 24] focused on increasing HPV vaccine uptake and completion rates by engaging preteens and their parents through the "*Land of Secret Gardens*" game. Two studies [22, 27] aimed to boost knowledge about diseases, HPV, cervical cancer, and vaccinations through educational content integrated into gameplay. Two studies [25, 26] aimed to influence perceptions and self-efficacy related to HPV, with one targeting college-aged men and the other focusing on children's knowledge and attitudes toward vaccination. Two studies [21, 26] aimed to promote positive attitudes toward vaccination by increasing understanding of HPV and encouraging vaccine uptake through educational games and multimedia content.

Components and reward system

The framework of gamification comprises various elements such as rewards, feedback mechanisms, motivation, challenges, and learning components (Fig. 3). Five studies incorporated educational content as a central component, supported by feedback mechanisms to reinforce learning [23, 24, 26–28]. Regarding rewards, physical character cards are used [21, 22], while three studies offered financial incentives such as gift cards and Visa debit cards [23–25]. Three studies motivated players with stars, scores, leaderboards, and achievements [26–28], while one study [21] combined virtual rewards with real-world incentives, including coupons for a free vaccine.

Attitude/Behavior outcomes

Five studies reported the impact of gamification intervention on HPV vaccination behaviors [24] presented in Table 2. A significant influence ($p < .001$) was found on the decision to receive the HPV vaccine, particularly when participants could customize their avatars to represent their ideal or actual selves [25]. Cate et al., 2018 [24] observed increased self-efficacy and confidence among preteens with the customization of avatars playing a crucial role in enhancing their commitment. This indicates that personalized elements can effectively drive vaccine-related behaviors. Hasni et al., 2023 [28] reported increased engagement and more positive attitudes toward the HPV vaccine through game personalization, suggesting that tailoring the gaming experience to users' preferences can favourably influence their attitudes. Additionally, Chau et al., 2023 [21] found improvements in both attitudes and confidence regarding vaccine safety, emphasizing that educational resources, including interviews with healthcare professionals, effectively enhanced participants' perceptions of the vaccine's safety and increased their openness to vaccination.

On the other hand, two studies reported no significant impact on attitudes toward the HPV vaccine [22] and found no change in consent rates between experimental and control groups.



Fig. 3 Key Elements in gamification intervention strategies used across the studies

Table 2 Changes in HPV vaccine knowledge, attitudes/behaviors, and HPV vaccination intentions following the gamification intervention

Study	Outcomes Measured	Theoretical framework	Findings of the study (knowledge, attitudes, behavior, vaccine intention related)	Attitude/Behaviour	Knowledge	Vaccine uptake/intention
[19]	Preteen engagement and preferences, knowledge, and awareness about HPV vaccination their attitudes toward receiving the vaccine.	The Health Belief Model	Preteens expressed a desire to protect themselves against the virus and "be safe and healthy in the future. Gamification increased self-efficacy and confidence on the HPV vaccine	Behavioral customizing an avatar to represent the ideal self being the most indicative of an increased level of behavioral intention to receive the HPV vaccine	Both preteens and parents wanted more information on HPV transmission, prevention, and vaccine safety, which the game effectively provided engagingly and memorably.	Not reported
[24]	The main outcome was the changes in knowledge, beliefs, attitudes, and communication	Not reported	The results indicate that the intervention had a significant positive effect on both the initiation and completion of the HPV vaccine series during and post-intervention periods.	Not reported	Not reported	The intervention significantly increased both initiation (HR = 1.17, p = .004) and completion rates (17% higher during the intervention, 30% higher post-intervention, p = .03), after adjusting for baseline differences.
[23]	HPV Vaccination knowledge or vaccination self-efficacy	Self-determination theory	The comparison group had a higher self-efficacy score than the intervention group (1.65 vs. 1.45; P = .05). While the overall mean decisional balance score favored vaccination, the differences between groups were not significant.	Not reported	Postintervention, the mean knowledge score (5 items, range 1–3) was higher in the intervention group than in the comparison group (2.56, SD 0.34 vs. 2.28, SD 0.41, respectively; P = .03)	Vaccine initiation and completion rates were higher in the intervention group (22% vs. 15%; P = .31) than in the comparison group (9% vs. 2%; P = .10), although the difference was not significant.
[25]	The main outcome was risk perceptions, self-efficacy and behavioral intention to receive the HPV vaccine	Health Belief Model (HBM) and Theory of Planned Behavior	Data analysis indicates an increase in gain score for risk perception, self-efficacy and behavioral intention when participants were able to customize their avatar to look like their ideal or actual self	No significant interaction was found between avatar type and self-perception ($F(3,104) = 1.549$, p = .206) or for condition group ($F(3,104) = 0.249$, p = .862). However, the experimental game with an avatar significantly influenced behavioral intention ($F(1,104) = 33.907$, p < .001).	Not reported	Not reported
[22]	vaccine consent rates (HPV) from 2018 to 2019 for each school and between the experimental and control groups	Theory of change	The study showed no significant change in overall consent rates. However, subgroup analysis revealed that students from small schools had a notable increase in consent to vaccination between 2018 and 2019.	Not reported	Not reported	No significant difference in consent rates was found between the experimental and control groups (p = .37, mean difference = −0.05, 95% CI: −0.16 to 0.07).

Table 2 (continued)

Study	Outcomes Measured	Theoretical framework	Findings of the study (knowledge, attitudes, behavior, vaccine intention related)	Attitude/Behaviour	Knowledge	Vaccine uptake/intention
[26]	attitudes, knowledge, subjective norms, self-efficacy, intentions, and emotions regarding the HPV vaccine	Theory of planned behavior	Both the animated video and the game significantly improved knowledge and intention regarding HPV. The video also increased self-efficacy and reduced fears, while the game did not impact self-efficacy or fears but did enhance intention.	No significant change in attitude was observed with the animated video (mean difference = 0.15, p =.42) or the game (mean difference = 0.08, p =.33).	Both the animated video and the game significantly improved knowledge scores after the intervention (animated video: mean difference = 0.55, p <.001; game: mean difference = 0.45, p <.001)	The game significantly increased intention scores (mean difference = 0.80, p =.01), while the animated video also showed a significant increase (mean difference = 0.60, p <.001)
[27]	HPV knowledge, attitude toward HPV vaccination, Player Engagement, User satisfaction, and experience	Social cognitive theories	The game effectively increased users' knowledge about HPV, as demonstrated by a significant improvement in HPV-related knowledge after playing.	Not reported	FightHPV significantly improved participants' knowledge from median score 2.00 (IQR: 1.00–2.00) before to 2.00 (IQR: 2.00–3.75.00.75) after (p =.006). Players understanding of HPV-related topics improved after interacting with the game.	Not reported
[28]	Engagement, awareness, and knowledge of the HPV vaccine and cervical cancer	Theory of Planned Behavior	Findings indicated that personalizing game characters and scenarios could boost engagement, with adolescents showing a preference for customized content based on their interests.	The study observed a high level of engagement among participants, reflecting their knowledge and awareness of social issues. High engagement can be indicative of positive changes in attitude towards the HPV vaccine,	The results showed that all participants were highly engaged with the "HPV vaccine" game, the boy character reflects the player's knowledge and awareness of social lives	Not reported
[21]	Participants' knowledge, attitudes, and behaviors related to HPV vaccination.	Not reported	The findings indicate that the resources were well-received and effective in increasing knowledge about HPV, dispelling misconceptions, and fostering positive attitudes toward vaccination.	The resources improved participants' attitudes toward HPV vaccination, with healthcare professionals' interviews boosting their confidence in the vaccine's safety and increasing their openness to vaccination.	The educational resources, including the Kahoot game, enhanced participants' understanding of HPV and its vaccination, covering vaccine efficacy, types, and the importance of early vaccination.	The study noted an increase in vaccination intention, particularly among participants who recognized the personal relevance and urgency of getting vaccinated.

Knowledge outcomes

Six studies reported a positive change in knowledge regarding HPV vaccination [17, 19, 22–25]. Preteens demonstrated improved vaccine initiation behaviors, attributed to an enhanced understanding of the HPV vaccination process [17]. Similarly [23], observed higher knowledge scores in the intervention group (2.56, SD 0.34 vs. 2.28, SD 0.41, p =.03) following the intervention, reinforcing the effectiveness of targeted educational strategies. Several studies emphasized the importance of interactive formats in enhancing knowledge. For instance [28], reported increased knowledge as a result of game engagement, while [26] highlighted a marked improvement in knowledge scores after participants engaged with both the video and the game (MD: 0.55, p <.001; game: MD: 0.45, p <.001). Similarly [27], found significant gains in HPV-related knowledge following game interaction (2.00, IQR: 2.00–3.75.00.75, p =.006). Additionally [21],

demonstrated an enhanced understanding of HPV and vaccination among participants, emphasizing the positive impact of educational resources. All included studies assessed knowledge change using pre- and post-intervention questionnaires, with the following exceptions [28] measured knowledge through a 15-item in-game Q&A across three levels (HPV, cervical cancer, and vaccine), while the qualitative study by [21] evaluated perceived and demonstrated knowledge change through participants' interview responses.

Vaccine uptake/Intention outcomes

Five studies reported a significant enhancement in HPV vaccine initiation [21–24, 26]. One study observed a notable increase in vaccine initiation and completion rates, with the intervention group showing a significant (p =.004) vaccine initiation intention, and a 17% higher completion rate during the intervention and a 30% higher rate post-intervention, both achieving statistical significance (p =.03) [24]. Additionally, the intervention significantly increased vaccine intention scores, with the game showing a mean difference of 0.80 (p =.01), and the animated video also yielded a significant increase (mean difference = 0.60, p <.001) [26]. Moreover, educational resources, including a Kahoot game, contributed to increased vaccination intention, particularly among participants who recognized the personal relevance of the vaccine [21].

However, despite these improvements, no significant difference in consent rates was observed between the experimental and control groups (p =.37, mean difference = −0.05, 95% CI: −0.16 to 0.07) [22]. Similarly, while higher completion rates were observed in the intervention group compared to the comparison group, these differences were not statistically significant [23]. This suggests that, although the intervention demonstrated a positive trend in enhancing vaccine initiation and intention, the effect on actual vaccine consent and completion remains less conclusive.

Discussion

This systematic review synthesized evidence from nine studies evaluating gamification interventions aimed at improving HPV vaccine-related knowledge, attitudes, intentions, and uptake among adolescents and young adults. Overall, the findings suggest that gamification can enhance knowledge and positively shape attitudes toward HPV vaccination, while also showing potential to increase vaccination intentions and initiation.

Gamification, the use of game-design elements in non-game contexts, has emerged as a promising tool to enhance health-related behaviors, including vaccination uptake. Knowledge gains emerged as the most consistent outcome across studies, underscoring the potential of gamification to address misinformation and limited awareness [29, 30], two of the most persistent barriers to HPV vaccination. Among various eHealth interventions, game-based approaches have shown promising outcomes in enhancing knowledge acquisition, retention, and promoting behavioral changes in youth [31, 32]. For promoting HPV vaccine uptake, gamification on digital platforms is widely used [19, 21, 23, 24, 26–28]. These digital games serve as powerful tools by immersing users in engaging scenarios where they learn about HPV, its transmission, and the benefits of vaccination in an interactive and enjoyable manner [33]. Given that the youth population is both tech-savvy and often influenced by peer dynamics, digital gamification can act as a bridge between knowledge acquisition and action, ultimately encouraging young people to make informed decisions about their health [17]. By incorporating digital games, gamification intervention taps into the preferences of a generation accustomed to digital experiences. For young people, who may otherwise be indifferent or skeptical about vaccinations, these elements can transform educational content into something dynamic and enjoyable, leading to greater engagement and understanding [17, 34].

Gamification interventions often include features like rewards, instant feedback, and progress tracking, which encourage users to engage more deeply by transforming dry, fact-based content into dynamic experiences that actively engage users, making the learning process enjoyable while also driving meaningful behavioral outcomes [35]. By integrating sustained engagement through elements like goal-setting, competition, rewards, and social interaction, storytelling, role-playing, and real-life scenarios into game designs, gamified interventions can challenge pre-existing negative beliefs or fears about vaccines, replacing them with more informed and positive attitudes [35, 36]. Gamification engages users by integrating rewards and incentives that offer immediate positive reinforcement. One of the key aspects of this approach is the use of points, badges, or levels that users earn as they complete tasks [37]. Another powerful element in gamification is the sense of progression. Users move through levels, gaining a sense of achievement as they advance. The HPV vaccination is administered in a two-dose schedule; completing both doses is crucial for achieving optimal protection against HPV-related diseases. Achieving small milestones fosters a sense of accomplishment, which in turn motivates users to reach the ultimate goal: completing all doses of the HPV vaccine [24]. Challenges and competition are key motivators in gamification, encouraging users to push themselves by competing with peers or communities, such as who learns the most about HPV or completes vaccination first. This competitive spirit, combined with camaraderie, boosts

vaccination rates [20]. Real-time feedback, through quizzes or modules, helps users track their progress, correct misconceptions, and stay engaged [19, 23, 24]. Personalization allows users to create avatars or engage with narratives that reflect their own lives. Stories embedded in games enhance self-efficacy by providing vicarious experiences, emotional arousal, and motivation to adopt desired health behaviors [22, 25–28]. The gaming principles, such as the actions of avatars—graphical representations of individuals within a virtual space—along with the narrative, facilitate observational learning [38]. This approach models effective strategies that lead to positive behaviors and illustrate the adverse consequences of undesirable actions, promoting a deeper understanding of health-related choices.

While gamification holds significant potential, it faces notable barriers, particularly in resource-limited regions. One of the major challenges is the digital divide, which refers to the unequal access to technology, internet connectivity, and digital devices in many parts of the world [39]. In low- and middle-income countries, where HPV vaccination efforts are critically needed, access to smartphones, tablets, or reliable internet is often limited. This restricts the reach of digital gamified interventions, preventing large segments of the population from benefiting from these innovative strategies [40]. In addition to technology access, digital literacy is essential for the successful implementation of gamification [41]. Users must be able to navigate and engage with digital platforms effectively, but in resource-limited areas, digital literacy may be lower. Without the skills to fully interact with gamified content, users may struggle to understand or engage, reducing the overall impact of these interventions. Another barrier is cultural relevance [42]. Gamified interventions developed without considering local beliefs, customs, or health perceptions may fail to resonate with the target audience. In some regions, there may be strong cultural resistance to vaccination or specific misconceptions about the HPV vaccine. If gamified tools don't address these concerns thoughtfully, they may struggle to influence behaviors. Tailoring games to be culturally sensitive and relevant to the local context is essential for overcoming these obstacles [42]. The cost of developing and maintaining high-quality gamified platforms is another significant challenge. Creating engaging digital games requires substantial investment in design, technology, and ongoing updates. In regions where healthcare budgets are already limited, allocating resources to develop and sustain gamified interventions might be difficult. Even if these tools are developed, they may not reach their full potential without adequate funding for implementation, distribution, and maintenance.

Limitations and future directions

One important limitation is that the gamification interventions for HPV vaccination have not completely matured, and a paucity of strong evidence of their effectiveness exists. Therefore, the number of studies investigating the gamification of HPV knowledge, behavior, attitude, or vaccine uptake is limited. Additionally, the studies that were included in this review lacked global representation, with very few data available for most continents and regions of the world, especially those most resource-poor. The majority of studies included were from various settings, but mostly developed countries; thus, elitism in this regard could bias the results (few), especially in developing countries. The complexities and understudied nature of the barriers to implementation of gamification interventions, or digital gaming in these regions, remain little understood. Future research should explore both digital and non-digital gamification strategies in LMIC contexts to address persistent gaps in HPV vaccine uptake. Furthermore, the studies included in this review lacked global representation, with very few data available for most continents and regions of the world, especially those most resource-poor. Nevertheless, although efficacy is limited, this review still provides a reference for further research, especially through RCTs, to show the effectiveness of gamification interventions in HPV vaccination programs. In addition, the included studies were highly heterogeneous in study design. Most are qualitative; few are quantitative studies or RCTs comparing gamification interventions to control groups. While a few RCTs are now underway, the need for future investigations using stricter methodologies, including larger sample sizes and more conceptually full approaches to evaluating the effectiveness of gamification in increasing HPV vaccine coverage, is evident. Future research is warranted to prioritize expanding the geographical and methodological scope of studies on gamification and HPV vaccination, with a particular focus on resource-limited settings and the use of RCTs to provide stronger evidence.

The strength of the evidence differs by outcome. Knowledge effects were moderate to strong and consistent, while intention and uptake outcomes showed heterogeneity. Some interventions demonstrated significant improvements in initiation and completion rates, while others reported null effects. These discrepancies may reflect differences in population (e.g., preteens vs. university students), intervention intensity, or contextual barriers such as parental consent. Intervention type also mattered: digital serious games with embedded feedback and reward systems tended to yield stronger results than non-digital formats. Importantly, the identification of intervention features—such as feedback loops, progress milestones, and reward mechanisms is an original

contribution of this review, offering actionable insights for future program design.

Conclusion

This systematic review demonstrates the promising potential of gamification interventions in enhancing HPV vaccine-related knowledge, attitudes, behaviors, and uptake intentions among adolescents and young adults. The diverse studies highlight effective strategies such as interactive and personalized game mechanics, educational content, and social engagement features. However, the variability in outcomes emphasizes the need for further research to optimize gamification components and ensure broader applicability across different populations and settings. The findings underscore the critical role of tailored, engaging platforms in public health interventions, with implications for scaling efforts to improve vaccine awareness and acceptance globally.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-025-25606-8.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Acknowledgements

The authors would like to acknowledge Safah Sohail for providing digital support in the creation of Figure 3 and refining the reference list

Author contributions

S.T . conceived the research idea, developed the study methodology, and prepared the protocol. **S.T.** also led the screening process and conducted the data analysis. In addition, **S.T.** wrote the main manuscript, finalized the references, and prepared Figures 1-3, as well as designed Tables 1-2, ensuring the accuracy and consistency of all visual elements. **S.T.** revised the manuscript as according the reviewer comments **R.A** . collaborated with **S.T.** in the screening of the studies, and **R.A** was also responsible for the editing of the manuscript. **G.N** . completed the proofreading of the manuscript.**M.T.Y.** supervised the entire project, offering guidance and oversight at all stages of the work.

Funding

Not applicable.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

[1]Department of Community Health and Science, Aga Khan University, Karachi, Pakistan
[2]Rehman Medical College, Peshawar, Pakistan
[3]Department of Public Health, Jinnah Sindh Medical University, Karachi, Pakistan
[4]School of Population Health, University of New South Wales, Sydney, Australia

Received: 27 June 2025 / Accepted: 7 November 2025
Published online: 11 February 2026

References

1. WHO. Human papillomavirus and cancer. 2024. https://www.who.int/news-room/fact-sheets/detail/human-papilloma-virus-and-cancer.
2. Bruni L, Albero G, Serrano B, Mena M, Collado JJ, Gómez D, Muñoz J, Bosch FX, de Sanjosé S. Human Papillomavirus and Related Diseases in the World. Summary Report 10 March 2023. ICO/IARC Information Centre on HPV and Cancer (HPV Information Centre); 2023. Available from: https://hpvcentre.net/statistics/reports/XWX.pdf.
3. Roman BR, Aragones A. Epidemiology and incidence of HPV-related cancers of the head and neck. J Surg Oncol. 2021;124(6):920–2.
4. Soheili M, Keyvani H, Soheili M, Nasseri S. Human papilloma virus: a review study of epidemiology, carcinogenesis, diagnostic methods, and treatment of all HPV-related cancers. Med J Islam Repub Iran. 2021;35:65.
5. Chaturvedi AK, Engels EA, Pfeiffer RM, Hernandez BY, Xiao W, Kim E, et al. Human papillomavirus and rising oropharyngeal cancer incidence in the United States. J Clin Oncol. 2023;41(17):3081–8.
6. Saraiya M, Unger ER, Thompson TD, Lynch CF, Hernandez BY, Lyu CW, et al. US assessment of HPV types in cancers: implications for current and 9-valent HPV vaccines. J Natl Cancer Inst. 2015;107(6):djv086.
7. Steben M, Tan Thompson M, Rodier C, Mallette N, Racovitan V, DeAngelis F, et al. A review of the impact and effectiveness of the quadrivalent human papillomavirus vaccine: 10 years of clinical experience in Canada. J Obstet Gynaecol Can. 2018;40(12):1635–45.
8. Hirth J. Disparities in HPV vaccination rates and HPV prevalence in the United States: a review of the literature. Hum Vaccin Immunother. 2019;15(1):146–55.
9. Reagan-Steiner S, Yankey D, Jeyarajah J, Elam-Evans LD, Singleton JA, Curtis CR, et al. National, regional, state, and selected local area vaccination coverage among adolescents aged 13–17 years–United States, 2014. MMWR Morb Mortal Wkly Rep. 2015;64(29):784–92.
10. Pingali CYD, Chen M, et al. National vaccination coverage among adolescents aged 13–17 years — National immunization survey-teen, United States, 2023. MMWR Morb Mortal Wkly Rep. 2024;73:708–14.
11. HHS.gov. Increase the proportion of adolescents who get recommended doses of the HPV vaccine — IID–08. In. 2023. https://odphp.health.gov/healthypeople/objectives-and-data/browse-objectives/vaccination/increase-proportion-adolescents-who-get-recommended-doses-hpv-vaccine-iid-08/data-methodology.
12. Sardi L, Idri A, Fernandez-Aleman JL. A systematic review of gamification in e-health. J Biomed Inform. 2017;71:31–48.
13. Hamari JKJ, Sarsa H. Does gamification work? - A literature review of empirical studies on gamification. In: Annual Hawaii International Conference on System Sciences. Manoa. 2014. https://doi.org/10.1109/HICSS.2014.377.
14. Johnson D, Deterding S, Kuhn KA, Staneva A, Stoyanov S, Hides L. Gamification for health and wellbeing: a systematic review of the literature. Internet Interv. 2016;6:89–106.
15. Lister C, West JH, Cannon B, Sax T, Brodegard D. Just a fad? Gamification in health and fitness apps. JMIR Serious Games. 2014;2(2):e9.
16. Hakim H, Driedger SM, Gagnon D, Chevrier J, Roch G, Dube E, et al. Digital gamification tools to enhance vaccine uptake: scoping review. JMIR Serious Games. 2024;12:e47257.
17. Montagni I, Mabchour I, Tzourio C. Digital gamification to enhance vaccine knowledge and uptake: scoping review. JMIR Serious Games. 2020;8(2):e16983.
18. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ. 2021;372:n71.

19. Cates JR, Crandell JL, Diehl SJa, Coyne-Beasley T. Immunization effects of a communication intervention to promote preteen HPV vaccination in primary care practices. Vaccine. 2018;36(1):122–7.
20. Darville G, Burns J, Chavanduka T, Anderson-Lewis C. Utilizing theories and evaluation in digital gaming interventions to increase human papillomavirus vaccination among young males: qualitative study. JMIR Serious Games. 2021;9(1):e21303.
21. Chau JPC, Lo SHS, Butt L, Lee VWY, Lui GCY, Lau AYL. User engagement on a novel educational health intervention aimed at increasing HPV vaccine uptake in Hong Kong: a qualitative study. J Cancer Educ. 2023;38(3):772–80.
22. Epstein D, Enticott J, Larson H, Barton C. Pragmatic cluster randomised control trial using vaxcards as an age-appropriate tool to incentivise and educate school students about vaccination. BMJ Open. 2021;11(9):e049562.
23. Cates JR, Fuemmeler BF, Stockton LL, Diehl SJ, Crandell JL, Coyne-Beasley T. Evaluation of a serious video game to facilitate conversations about human papillomavirus vaccination for preteens: pilot randomized controlled trial. JMIR Serious Games. 2020;8(4):e16883.
24. Cates JR, Fuemmeler BF, Diehl SJ, Stockton LL, Porter J, Ihekweazu C, et al. Developing a serious videogame for preteens to motivate HPV vaccination decision making: land of secret gardens. Games Health J. 2018;7(1):51–66.
25. Darville S, Burns J, Chavanduka T, Anderson-Lewis C. The male Domain-Digital Game-Based learning for human papillomavirus vaccination among young males. Games Health J. 2022;11(5):312–20.
26. Occa A, Stahl HM, Julien-Bell S. Helping children to participate in human Papillomavirus-related discussions: mixed methods study of multimedia messages. JMIR Form Res. 2022;6(4):e28676.
27. Ruiz-Lopez T, Sen S, Jakobsen E, Trope A, Castle PE, Hansen BT, et al. FightHPV: design and evaluation of a mobile game to raise awareness about human papillomavirus and nudge people to take action against cervical cancer. JMIR Serious Games. 2019;7(2):e8540.
28. Hasni N, Azalian A, Kadir T, Hisham MAM. Serious human papillomavirus vaccine game for Malaysian adolescents: development and preliminary study. Int J Public Health Sci. 2013;12(1):82–90.
29. Chen X, Wang L, Huang Y, Zhang L. Risk perception and trust in the relationship between knowledge and HPV vaccine hesitancy among female university students in China: a cross-sectional study. BMC Public Health. 2024;24(1):667.
30. Chowdhury S, Ara R, Roy S, Tanvir SMS, Eva FN, Neela TM, et al. Knowledge, attitude, and practices regarding human papillomavirus and its' vaccination among the young medical professionals and students of Bangladesh. Clin Exp Vaccine Res. 2022;11(1):63–71.
31. Amresh A, Karnick P. Creating an interest in computer graphics by teaching game development. Eurographics Educ Papers. 2007. p. 9–16.
32. Baranowski T, Buday R, Thompson D, Lyons EJ, Lu AS, Baranowski J. Developing games for health behavior change: getting started. Games Health J. 2013;2(4):183–90.
33. Warren JR, Hopfer S, Fields EJ, Natarajan S, Belue R, McKee FX, et al. Digital HPV education to increase vaccine uptake among low income women. PEC Innovation. 2023;2:100111.
34. Ou L, Reifsnider E, Chen AC, Todd M, Amresh A, Mun CJ. Using gaming to promote vaccination among youth: a systematic review. SAGE Open Nursing. 2024;10:23779608241256998.
35. Hellín C, Calles-Esteban F, Valledor A, Gómez J, O-TS, Tayebi A. Enhancing student motivation and engagement through a gamified learning environment. Sustainability. 2023;15(19):14119.
36. Gkintoni E, Vantaraki F, Skoulidi C, Anastassopoulos P, Vantarakis A. Promoting physical and mental health among children and adolescents via gamification-a conceptual systematic review. Behav Sci (Basel). 2024. https://doi.org/10.3390/bs14020102.
37. Alsawaier RS. The effect of gamification on motivation and engagement. Int J Inform Learn Technol. 2018;35(1):56–79.
38. Rodrigues L, Palomino PT, Toda AM, Klock AC, Oliveira W, Avila-Santos AP, Isotani S. Personalization improves gamification: evidence from a mixed-methods study. In: Proceedings of the ACM on Human-Computer Interaction: 2021. 2021. p. 1–25.
39. Sanchez-Mena A, Marti-Parreno J. Drivers and barriers to adopting gamification: teachers' perspectives. Electron J E-Learning. 2017;15:434–43.
40. Taseen S, Yousafzai MFHQ. Harnessing artificial intelligence and digital technology for enhancing routine immunization among zero-dose children. Digit Health. 2025;11:1–12.
41. Kaimara P, Fokides E, Oikonomou A, Deliyannis I. Potential barriers to the implementation of digital game-based learning in the classroom: pre-service teachers' views. Technol Knowl Learn. 2021;26(4):825–44.
42. Naderbagi A, Loblay V, Zahed IUM, Ekambareshwar M, Poulsen A, Song YJC, et al. Cultural and contextual adaptation of digital health interventions: narrative review. J Med Internet Res. 2024;26:e55130.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.